

23 from the Gradient Vectors

$$\nabla f(x, y) = f_x \mathbf{i} + f_y \mathbf{j} = x\mathbf{i} + y\mathbf{j}$$

4. Evaluate the gradient vector at the point (1, 1)

$$\nabla f(x, y) = f_x \mathbf{i} + f_y \mathbf{j} = x\mathbf{i} + y\mathbf{j}$$

$$\nabla f(1, 1) = (1)\mathbf{i} + (1)\mathbf{j} = \mathbf{i} + \mathbf{j}$$

(a) Direction of Most Rapid increase (Max Rate)

The function increases most rapidly in the direction of the positive gradient vector, $\nabla f(1, 1)$. To define a direction, we use the unit vector ($\mathbf{u}$) in that direction.

→ Find the magnitude of $\nabla f(1, 1)$:

$$= \sqrt{1^2 + 1^2} = \sqrt{2}$$

3. calculate the Unit Vector ($\mathbf{u}$)

$$\mathbf{u} = \frac{\mathbf{i} + \mathbf{j}}{\sqrt{2}} = \frac{1}{\sqrt{2}} \mathbf{i} + \frac{1}{\sqrt{2}} \mathbf{j}$$

(b) Direction of Most Rapid Decrease (Min Rate)

→ The function decrease most rapidly in the direction opposite to the gradient vector, which is $-\nabla f(1,1)$

The direction is the negative unit vector

$-u$:

$$-u = -\left(\frac{1}{\sqrt{2}}i + \frac{1}{\sqrt{2}}j\right) = -\frac{1}{\sqrt{2}}i - \frac{1}{\sqrt{2}}j$$

(c) Directions of Zero change

The function has zero change (it's flat) when moving in any direction (orthogonal, perpendicular)

to the gradient vector. $\nabla f(1,1)$, if $\nabla f =$

$ai + bj$, the two unit vectors n orthogonal to it are found by swapping the components and changing the sign of one them.

1. Gradient vector $= \nabla f(1,1) = 1i + 1j$

2. Orthogonal vector $m =$ swap components and change the sign of the i component
 $-1i + 1j$

3. Unit vectors of zero change = we divide this vector by its magnitude, which

□

$$\sqrt{(-1)^2 + (1)^2} = \sqrt{2}$$

$$n = \frac{-1i + 1j}{\sqrt{2}} = \frac{-1}{\sqrt{2}} i + \frac{1}{\sqrt{2}} j$$

therefore,

$$n = \frac{-1}{\sqrt{2}} i + \frac{1}{\sqrt{2}} j \text{ and } -n = \frac{1}{\sqrt{2}} i - \frac{1}{\sqrt{2}} j$$